# 5.3 Principles of Electrochemistry (A Level Only)

## Question Paper

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|---|---|
| Course | CIE A Level Chemistry |
| Section | 5. Physical Chemistry (A Level Only) |
| Topic | 5.3 Principles of Electrochemistry (A Level Only) |
| Difficulty | Easy |

**Time allowed:** 50

**Score:** /39

**Percentage:** /100

## Question 1a

This question is about electrolysis.

Explain what is meant by the term electrolysis.

**[2 marks]**

## Question 1b

Complete Table 1.1 to show the correct charge associated with each term.

**Table 1.1**

| Term | Charge (positive / negative) |
|---|---|
| Anion | |
| Anode | |
| Cathode | |
| Cation | |

**[4 marks]**

## Question 1c

i)

Explain what reduction and oxidation mean in terms of electrons.

[1]

ii)

Write a chemical equation for the reduction of copper(II) ions, $Cu^{2+}$ to form copper metal.

[1]

iii)

Complete the chemical equation for the oxidation of chloride ions, $Cl^{-}$ to form chlorine, $Cl_2$.

$$2Cl^{-} \rightarrow \text{.........} + \text{.........}$$

[1]

**[3 marks]**

## Question 2a

Electrolysis is used to extract reactive metals from their metal ores, purify metals and produce non-metals such as fluorine.

Suggest why ionic compounds need to be molten or aqueous for electrolysis to occur.

**[1 mark]**

## Question 2b

i)

Complete Table 2.1 to show the electrolysis products of the molten ionic compounds.

**Table 2.1**

| Ionic compound | Product at anode | Product at cathode |
|---|---|---|
| NaH | Hydrogen | Sodium |
| $CaCl_2$ | | |
| $Pb_3O_4$ | | |

[2]

ii)

State why sodium forms at the negative cathode during the electrolysis of sodium hydride, NaH.

[1]

**[3 marks]**

## Question 2c

Electrolysis of aqueous ionic solutions can give different products compared to the electrolysis of molten ionic compounds.

Give the formulae of **two** ions that can form the different products during the electrolysis of aqueous solutions.

**[1 mark]**

## Question 2d

Two factors that affect the actual ions discharged during the electrolysis of aqueous solutions are:

- The relative electrode potential of the ions
- The concentration of the ions

i)

State the relationship between the relative electrode potential of the ions and the ions that are discharged during the electrolysis of an aqueous solution.

[1]

ii)

Describe the relationship between the concentration of the ions and the ions that are discharged during the electrolysis of an aqueous solution.

[1]

**[2 marks]**

## Question 3a

This question is about the electrolysis of ionic compounds and associated calculations.

State the equation linking charge, current and time. Your answer should include the units for each term.

Equation: ........................................

Units for charge: ..............................

Units for current: ...............................

Units for time: ....................................

**[2 marks]**

## Question 3b

The relationship between the Faraday constant, $F$, and Avogadro's constant, $L$, is shown in the equation.

$$F = L \times e$$

State what one Faraday measures.

**[1 mark]**

## Question 3c

Complete Table 3.1 to show the number of moles of electrons and the amount of charge for each equation.

**Table 3.1**

| Equation | Number of moles of electrons | Amount of charge / C |
|---|---|---|
| $K^+ + e^- \rightarrow K$ | 1 | 96 500 |
| $Cr^{3+} + 3e^- \rightarrow Cr$ | | |
| $S^{2-} \rightarrow$ ........ + ........ | | |

**[3 marks]**

## Question 3d

The chromium half-cell containing Cr / $Cr^{3+}$ is connected to a copper half-cell containing Cu / $Cu^{2+}$.

The standard electrode potential for copper is shown as:

$$Cu^{2+} + 2e^{-} \rightleftharpoons Cu \quad E^{\theta} = +0.34\ V$$

The standard cell potential of this combination was found to be + 1.08 V, where the copper half-cell would undergo reduction.

Calculate the standard electrode potential of the Cr / $Cr^{3+}$ half cell.

$E^{\theta}$ = .......... V

**[1 mark]**

## Question 3e

Write **two** half-equations to show what would occur if the chromium half-cell was attached to the standard hydrogen half-cell.

**[2 marks]**

## Question 4a

Standard electrode potentials are a measure of how easily a substance is reduced.

Give the standard conditions for the measurement of standard electrode potentials.

Temperature = ..............................

Pressure = ....................................

Ion concentration = ........................

**[3 marks]**

## Question 4b

Table 4.1 shows three standard electrode potentials.

**Table 4.1**

| | Equilibrium equation | Standard electrode potential, $E^{\theta}$ / V |
|---|---|---|
| **A** | $Ag^{+} + e^{-} \rightleftharpoons Ag$ | + 0.80 |
| **B** | $ClO^{-} + H_2O + 2e^{-} \rightleftharpoons Cl^{-} + 2OH^{-}$ | + 0.89 |
| **C** | $2H_2O + 2e^{-} \rightleftharpoons H_2 + 2OH^{-}$ | – 0.83 |

Identify the standard electrode potential with the equilibrium position that lies:

i)

Furthest left.

[1]

ii)

Furthest right.

[1]

**[2 marks]**

## Question 4c

The standard hydrogen electrode is used as the primary reference electrode for measuring standard electrode potential.

i)

Describe a standard hydrogen electrode.

[2]

ii)

State the standard electrode potential of the standard hydrogen electrode.

$E^{\theta}$ = .......... V

[1]

**[3 marks]**

## Question 5a

Electrochemical cells are made by connecting two half-cells with a salt bridge and a high-resistance voltmeter.

The standard hydrogen electrode is an example of a non-metal / non-metal ion half-cell which requires a platinum electrode.

Complete Table 5.1 to identify the **two** other types of half-cell and whether they require a platinum electrode.

**Table 5.1**

| Type of half-cell | Require a platinum electrode? |
|---|---|
| non-metal / non-metal ion | Yes |
| | |
| | |

**[2 marks]**

## Question 5b

The direction of electron flow can be determined by comparing the $E^{\theta}$ values of the two half-cells in an electrochemical cell.

In terms of poles, state the flow of electrons.

from .................................................. to ..................................................

**[1 mark]**

## Question 5c

Table 5.2 shows six standard electrode potentials.

**Table 5.2**

| Electrode reaction | $E^{\theta}$ / V |
|---|---|
| $Cu^{+} + e^{-} \rightleftharpoons Cu$ | + 0.52 |
| $Cu^{2+} + 2e^{-} \rightleftharpoons Cu$ | + 0.34 |
| $Cu^{2+} + e^{-} \rightleftharpoons Cu^{+}$ | + 0.15 |
| $Fe^{2+} + 2e^{-} \rightleftharpoons Fe$ | – 0.44 |
| $Fe^{3+} + 3e^{-} \rightleftharpoons Fe$ | – 0.04 |
| $Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$ | + 0.77 |

i)

Identify an electrode reaction that will ensure that the $Cu^{+}$ / Cu half-cell undergoes oxidation.

[1]

ii)

Write the equation for the feasible reaction of this electrochemical cell.

[2]

**[3 marks]**

# Model Answers: Easy

1a

a) Electrolysis means:

- Breaking a compound down into its elements; [1 mark]
- Using an electric current
  **OR**
  Using electricity; [1 mark]

**[Total: 2 marks]**

- This is a definition that you should know from your previous studies
- The language of the word electrolysis can help:
  - Electro - using electricity, although saying using an electric current is a stronger answer
  - Lysis - originally from the Greek word lysis meaning a loosening / setting free / releasing / dissolution / means of letting loose

1b

b) The completed table is:

| Term | Charge (positive / negative) |
|---|---|
| Anion | Negative; [1 mark] |
| Anode | Positive; [1 mark] |
| Cathode | Negative; [1 mark] |
| Cation | Positive; [1 mark] |

**[Total: 4 marks]**

- **Careful:** Examiners often comment about how these terms are not well understood
- For the electrodes, it can be helpful to remember PANIC:
  - **P**ositive **A**node, **N**egative **I**s **C**athode
- The ions, anion and cation, then have the appropriate charges to be attracted to their

related electrodes
- Anions are negative because they are attracted to the positive anode
- Cations are positive because they are attracted to the negative cathode

1c

c)

i) In terms of electrons, reduction and oxidation mean:

- Reduction = gain (of electrons)
  **AND**
  Oxidation = loss (of electrons); [1 mark]

ii) The chemical equation for the reduction of copper(II) ions, $Cu^{2+}$ is:

- $Cu^{2+} + 2e^{-} \rightarrow Cu$; [1 mark]

iii) The completed chemical equation for the oxidation of chloride ions, $Cl^{-}$ to form chlorine is:

- $2Cl^{-} \rightarrow Cl_2 + 2e^{-}$; [1 mark]

**[Total: 3 marks]**

- From your previous studies, you should be able to define reduction and oxidation in terms of electrons
  - **Remember:** OILRIG = oxidation is loss, reduction is gain
- Parts (ii) and (iii) are using chemical equations to check that you understand reduction and oxidation in terms of electrons
- In part (ii)
  - You are told that the copper(II) ions are $Cu^{2+}$ and are being reduced
  - This means that they must gain 2 electrons
- In part (iii)
  - You are told that the reaction is oxidation, which means that the chloride ions are losing electrons
  - Each chloride ion must lose one electron as it forms $Cl_2$

2a

a) Ionic compounds need to be molten or aqueous for electrolysis to occur because:

- (When aqueous or molten) they have mobile ions / ions that are free to move
  **OR**
  (When solid) there are no mobile ions
  **OR**
  (When solid) the ions are fixed in place / in fixed positions; [1 mark]

**[Total: 1 mark]**

- Electrolysis is the breakdown of an ionic compound using an electric current
- For the current to flow, the ions of the ionic compound **must** be able to move
  - **Careful:** If you talk about electrons moving, you will not get the mark

2b

b)

i) The completed table is:

| Ionic compound | Product at anode | Product at cathode | |
|---|---|---|---|
| NaH | Hydrogen / $H_2$ | Sodium / Na | |
| $CaCl_2$ | Chlorine / $Cl_2$ | Calcium / Ca | ; [1 mark] |
| $Pb_3O_4$ | Oxygen / $O_2$ | Lead / Pb | ; [1 mark] |

ii) Sodium forms at the cathode during the electrolysis of sodium hydride, NaH, because:

- Sodium is the cation / positive ion which is attracted to the cathode / negative electrode
  **OR**
  $Na^+$ is attracted to the cathode / negative electrode; [1 mark]

**[Total: 3 marks]**

- **Careful:** The question states that all of the ionic compounds are molten
- This means that the electrolysis products will be the component elements
- Do not be put off by the use of sodium hydride in part (ii), the component ions of sodium hydride are $Na^+$ and $H^-$
  - Since the cathode is negative, the positive sodium ion / $Na^+$ will be attracted there

2c

c) The formulae of **two** ions that can form the different products during the electrolysis of aqueous solutions are:

- $H^+$
  **AND**
  $OH^-$; [1 mark]

**[Total: 1 mark]**

- The key difference between the electrolysis of aqueous ionic solutions and molten ionic compounds is the water used to make the solution
- Water contains the hydrogen ion / $H^+$ and the hydroxide ion / $OH^-$
- Therefore, these ions must be responsible for the different products that can be formed

2d

d)

i) The relationship between the relative electrode potential of the ions and the ions that are discharged during the electrolysis of an aqueous solution is:

- The ion / cation with the most positive electrode potential / $E^\theta$ is discharged at the cathode / negative electrode
  **OR**
  The ion / anion with the most negative electrode potential / $E^\theta$ is discharged at the anode / positive electrode; [1 mark]

ii) The relationship between the concentration of the ions and the ions that are discharged during the electrolysis of an aqueous solution is:

- As the concentration of an ion increases, it is more likely to be discharged; [1 mark]

**[Total: 2 marks]**

- The command word for each part of this question is different
  - This is because the relationship between the relative electrode potential of the ions and the ions that are discharged is a clear fact

        - The ion with the most positive electrode potential / $E^\theta$ is discharged at the cathode
    - The relationship between the concentration of the ions and the ions that are discharged is not a clear fact
        - In a concentrated solution of NaF there are far more fluoride ions
            - Therefore, fluorine is discharged at the anode
        - In a very dilute solution of NaF there are more hydroxide ions than fluoride ions
            - Therefore, oxygen is discharged at the anode

3a

a) The equation linking charge, current and time is:

- $Q = I \times t$
  **OR**
  $I = \frac{Q}{t}$
  **OR**
  $t = \frac{Q}{I}$; [1 mark]

The units for each term are:

- Units for charge = coulombs
  **AND**
  Units for current = amps / amperes
  **AND**
  Units for time = seconds; [1 mark]

**[Total: 2 marks]**

Units for charge = coulombs
**AND**
Units for current = amps / amperes
**AND**
Units for time = seconds; [1 mark]

**[Total: 2 marks]**

- Questions using Faraday's Law are not very common but you should know the associated equations and be able to apply them

3b

b) One Faraday measures:

- The amount of (electric) charge carried by 1 mole of electrons / 1 mole of single charged ions; [1 mark]

**[Total: 1 mark]**

- You can also determine what a Faraday measures using the equation
  - *L* is Avogadro's constant which means that you are talking about one mole
  - *e* represents the charge on an electron

3c

c) The completed table is:

| **Equation** | **Number of moles of electrons** | **Amount of charge / C** |
|---|---|---|
| $K^+ + e^- \rightarrow K$ | 1 | 96 500 |
| $Cr^{3+} + 3e^- \rightarrow Cr$ | 3 | 289 500 |
| $S^{2-} \rightarrow S + 2e^-$ | 2 | 193 000 |

Chromium ion / $Cr^{3+}$ equation:

- Number of moles of electrons = 3
  **AND**
  Amount of charge = 289 500 (C); [1 mark]

Sulfur ion / $S^{2-}$ equation:

- $S^{2-} \rightarrow S + 2e^-$; [1 mark]
- Number of moles of electrons = 2
  **AND**
  Amount of charge = 193 000 (C); [1 mark]

**[Total: 3 marks]**

- The example line for the potassium ion / $K^+$ shows that the ion gains one mole of electrons, according to the equation
  - This uses 96 500 C of charge
- Applying this to the chromium ion / $Cr^{3+}$ equation:
  - There are 3 moles of electrons required (+ $3e^-$)
  - This uses 3 x 96 500 = 298 500 C of charge
- Applying the example line to the sulfur ion / $S^{2-}$ equation:
  - The sulfur ion will lose two moles of electrons to form sulfur
    - $S^{2-} \rightarrow S + 2e^-$
  - This uses 2 x 96 500 = 193 000 C of charge

3d

d) The standard electrode potential for the chromium half cell is:

- $E^\theta_{oxidised} = E^\theta_{reduced} - E^\theta_{cell} = 0.34 - 1.08 = -0.74$ (V); [1 mark]

**[Total: 1 mark]**

- This is a rearrangement of $E^\theta_{cell} = E^\theta_{reduced} - E^\theta_{oxidised}$
- It is vital to know whether the copper half-cell is the one showing oxidation or reduction to get this question right!

3e

e) The two half-equations are:

- $Cr (s) \rightarrow Cr^{3+} (aq) + 3e^-$; [1 mark]
- $2H^+ (aq) + 2e^- \rightarrow H_2 (g)$; [1 mark]

**[Total: 2 marks]**

- The standard electrode potential of Cr/ $Cr^{3+}$ is a negative value, so this will be oxidised
- The hydrogen half-cell will show reduction, which involves the gain of electrons by $H^+$

4a

a) The standard conditions for the measurement of standard electrode potentials are:

- Temperature = 298 K; [1 mark]
- Pressure = 1 atmosphere
  **OR**
  101 kPa; [1 mark]
- Ion concentration = 1.0 mol $dm^{-3}$; [1 mark]

**[Total: 3 marks]**

- You are expected to know the standard conditions as they don't just apply to electrochemistry

4b

b)

i) The standard electrode potential with the equilibrium position that lies furthest left is:

- **C** / $2H_2O + 2e^- \rightleftharpoons H_2 + 2OH^-$; [1 mark]

ii) The standard electrode potential with the equilibrium position that lies furthest right is:

- **B** / $ClO^- + H_2O + 2e^- \rightleftharpoons Cl^- + 2OH^-$; [1 mark]

**[Total: 2 marks]**

- When a standard electrode potential is positive, the equilibrium position lies to the right
  - The more positive the standard electrode potential value, the further right the equilibrium position
- When a standard electrode potential is negative, the equilibrium position lies to

the left
  - The more negative the standard electrode potential value, the further left the equilibrium position

4c

c)

i) The standard hydrogen electrode is:

- Hydrogen gas (at a pressure of 1 atm) bubbled into / in equilibrium with 1.00 mol $dm^{-3}$ solution of $H^+$ ions; [1 mark]
- With an inert platinum electrode (that is in contact with the hydrogen gas and $H^+$ ions); [1 mark]

ii) The standard electrode potential of the standard hydrogen electrode is:

- $E^{\theta}$ = 0 V; [1 mark]

**[Total: 3 marks]**

- It is more likely that you would be asked to draw the standard hydrogen electrode
- But, you need to know that:
  - Hydrogen gas is bubbled into a 1.00 mol $dm^{-3}$ solution of $H^+$ ions
    - This is particularly important when working with diprotic acids ($H_2SO_4$) and tripotic acids ($H_3PO_4$)
  - The electrode is made of platinum
- By definition, the standard hydrogen electrode has a standard electrode potential of 0.00 V

5a

a) The completed table is:

| Type of half-cell | Require a platinum electrode? | |
|---|---|---|
| non-metal / non-metal ion | Yes | |
| metal / metal ion | No | ; [1 mark] |

| | | |
|---|---|---|
| ion / ion | Yes | ; [1 mark] |

**[Total: 2 marks]**

- You should be aware of the three types of half-cell
- The metal / metal ion half-cell does not require a platinum electrode as the metal itself is used as the electrode
  - The non-metal / non-metal ion and ion / ion half-cells both require a platinum electrode

5b

b) In terms of poles, the flow of electrons is:

- From the negative pole to the positive pole; [1 mark]

**[Total: 1 mark]**

- This is an uncommon question
- However, when questions about the direction of electron flow are asked, examiners often comment on how it is not very well known / understood

5c

c)

i) The electrode reaction that will ensure that the $Cu^{+}$ / Cu half-cell undergoes oxidation is:

- $Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$; [1 mark]

ii) The equation for the feasible reaction of this electrochemical cell is:

| Type of half-cell | Require a platinum electrode? | |
|---|---|---|
| non-metal / non-metal ion | Yes | |
| metal / metal ion | No | ; [1 mark] |
| ion / ion | Yes | ; [1 mark] |

**[Total: 2 marks]**

- You should be aware of the three types of half-cell
- The metal / metal ion half-cell does not require a platinum electrode as the metal itself is used as the electrode
    - The non-metal / non-metal ion and ion / ion half-cells both require a platinum electrode

5b

b) In terms of poles, the flow of electrons is:

- From the negative pole to the positive pole; [1 mark]

**[Total: 1 mark]**

- This is an uncommon question
- However, when questions about the direction of electron flow are asked, examiners often comment on how it is not very well known / understood

5c

c)

i) The electrode reaction that will ensure that the $Cu^{+}$ / Cu half-cell undergoes oxidation is:

- $Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$; [1 mark]

ii) The equation for the feasible reaction of this electrochemical cell is:

$Fe^{3+} + Cu \rightleftharpoons Fe^{2+} + Cu^{+}$

- $Fe^{3+}$ reactant

**AND**

$Fe^{2+}$ product; [1 mark]

- Cu reactant

  **AND**

  $Cu^{+}$ product; [1 mark]

**[Total: 3 marks]**

- For part (i)
  - The $Cu^{+}$ / Cu half-cell to be the oxidation reaction, it must have the lowest standard electrode potential
    - The only electrode reaction that has a higher standard electrode potential is $Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$
- For part (ii)
  - The $Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$ electrode reaction / equation is written the correct way around as it is the reduction reaction
    - $Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$
  - The $Cu^{+} + e^{-} \rightleftharpoons Cu$ electrode reaction / equation has to be reversed so that it is the oxidation reaction
    - $Cu \rightleftharpoons Cu^{+} + e^{-}$
  - There is no need to multiply either equation by a coefficient because they both involve one electron
  - The two electrode reactions / equations are then combined
    - $Fe^{3+} + e^{-} + Cu \rightleftharpoons Fe^{2+}\ Cu^{+} + e^{-}$
  - The electrons cancel out to give the overall equation for the feasible reaction
    - $Fe^{3+} + Cu \rightleftharpoons Fe^{2+}\ Cu^{+}$